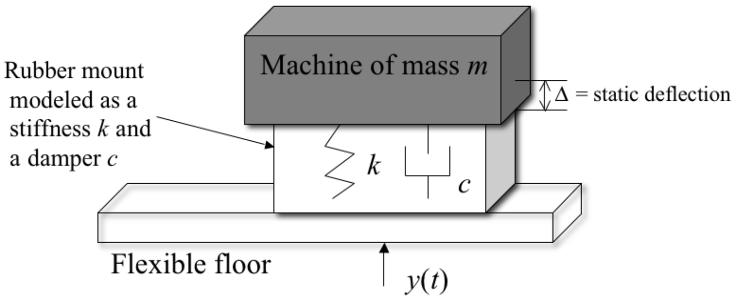
202010804 이경은

**2.54** A machine weighing 2000 N rests on a support as illustrated in Figure. The support deflects about 5 cm as a result of the weight of the machine. The floor under the support is somewhat flexible and moves, because of the motion of a nearby machine, harmonically near resonance ($r$ =1) with an amplitude of 0.2 cm. Model the floor as base motion, and assume a damping ratio of $z$= 0.01, and calculate the transmitted force and the amplitude of the transmitted displacement.




**Figure**

$$\frac{X}{Y} = \sqrt{\frac{1+(2\zeta r)^2}{(1-r^2)^2+(2\zeta r)^2}}$$

$$\frac{F_T}{kY} = r^2\sqrt{\frac{1+(2\zeta r)^2}{(1-r^2)^2+(2\zeta r)^2}}$$

$$mg - k\Delta = 0$$

$$k = \frac{mg}{\Delta} = 4\times10^4\ \mathrm{N/m}$$

$$Mg = 2000\mathrm{N}$$

$$m = 203\,\mathrm{kg}$$

$$\zeta = 0.01$$

$r=1$ 일때 $\quad Y = 0.2\times10^{-2} = 2\times10^{-3}$

$$\therefore X = Y\sqrt{\frac{1+(2\zeta r)^2}{(1-r^2)^2+(2\zeta r)^2}}$$

$$= 2\times10^{-3}\times\sqrt{\frac{1+4\times10^{-4}}{0+4\times10^{-4}}}$$

$$= 0.1\ m$$

$$F_T = kY r^2\sqrt{\frac{1+(2\zeta r)^2}{(1-r^2)^2+(2\zeta r)^2}}$$

$$= 4\times10^4\times2\times10^{-3}\sqrt{\frac{1+4\times10^{-4}}{0+4\times10^{-4}}}$$

$$= 4000\mathrm{N}$$

**3.12** Compute the response of the system:

$$3\ddot{x}(t) + 12\dot{x}(t) + 12x(t) = 3\delta(t)$$

subject to the initial conditions $x(0)$ = 0.01 m and $v(0)$ = 0. The units are in Newtons.

$\ddot{x} + 4\dot{x} + 4x = \delta(t)$

$\omega_n = 2$

$\zeta = 1$

$x_h = A_1 e^{-t} + A_2 t e^{-t}$

$= e^{-t}(A_1 + A_2 t)$

$\dot{x}_h = -e^{-t}(A_1 + A_2 t) + A_2 e^{-t}$

$x_0 = 0.01$

$v_0 = 0$

$A_1 = 0.01$

$-A_1 + A_2 = 0$

$\therefore x_h = 0.01 e^{-t}(t+1)$

$x_p = B_1 e^{-t} + B_2 t e^{-t}$

$x_0 = 0$

$v_0 = 1$

$= e^{-t}(B_1 + B_2 t)$

$\dot{x}_p = -e^{-t}(B_1 + B_2 t) + B_2 e^{-t}$

$B_1 = 0$

$+B_2 = 1$

$B_2 = 1$

$x(t) = 0.01 e^{-t}(t+1) + te^{-t} = 0.01 e^{-t} + 1.01 te^{-t}$

**3.23** A wave consisting of the wake from a passing boat impacts a seawall. It is desired to calculate the resulting vibration. Figure illustrates the situation and suggests a model. This force in Figure can be expressed as

$$F(t)=\begin{cases} F_0\left(1-\dfrac{t}{t_0}\right) & 0\le t\le t_0 \\ 0 & t>t_0 \end{cases}$$

Calculate the response of the seal wall-dike system to such a load.

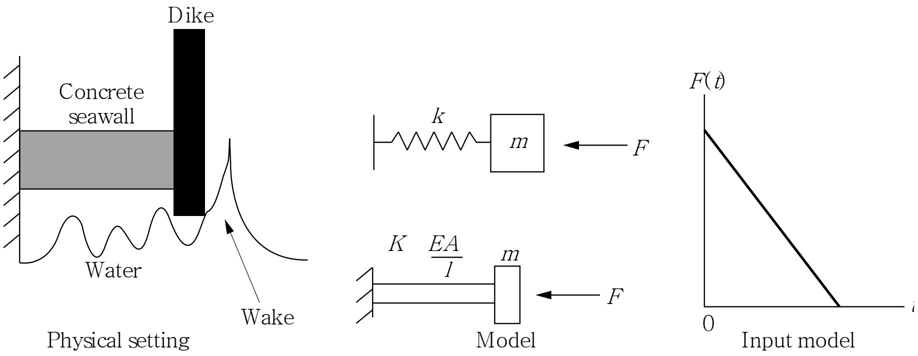



**Figure** Model of a wave hitting a dike.

$x_0=0,\ v_0=0$ 이라고 하면

$x_h=0$

$0<t<t_0$

$$x_p=\int_0^t F_0\left(1-\frac{\tau}{t_0}\right)\times\frac{1}{mw_n}\sin w_n(t-\tau)\,d\tau$$

$$=\frac{F_0}{mw_n}\int_0^t\left(1-\frac{\tau}{t_0}\right)\sin w_n(t-\tau)\,d\tau$$

$$=\frac{F_0}{mw_n}\left\{\left(1-\frac{\tau}{t_0}\right)\cos w_n(t-\tau)\Big|_0^t+\int_0^t\frac{1}{t_0}\cos w_n(t-\tau)\,d\tau\right\}$$

$$=\frac{F_0}{mw_n}\left\{\left(1-\frac{\tau}{t_0}\right)\cos w_n(t-\tau)\Big|_0^t-\frac{1}{w_nt_0}\sin w_n(t-\tau)\Big|_0^t\right\}$$

$$=\frac{F_0}{mw_n}\left\{\left(1-\frac{t}{t_0}\right)-\cos w_nt\quad+\frac{1}{w_nt_0}W_nt\right.$$

$$\therefore x=x_p=\frac{F_0}{mw_n}\left(1-\frac{t}{t_0}-\cos w_nt+\frac{1}{w_nt_0}\sin w_nt\right)\quad 0<t\le t_0$$

$t > t_0$

$$x_p = \int_0^{t_0} F_0\left(1-\frac{\tau}{t_0}\right)\frac{1}{m\omega_n}\sin\omega_n(t-\tau)\,d\tau + \int_{t_0}^{t} 0 \cancelto{0}{\times h(t-\tau)}\,d\tau$$

$$= \frac{F_0}{m\omega_n}\left\{\left(1-\frac{\tau}{t_0}\right)\cos\omega_n(t-\tau)\Big|_0^{t_0} - \frac{1}{\omega_n t_0}\sin\omega_n(t-\tau)\Big|_0^{t_0}\right\}$$

$$= \frac{F_0}{m\omega_n}\left(-\cos\omega_n t - \frac{1}{\omega_n t_0}\sin\omega_n(t-t_0) + \frac{1}{\omega_n t_0}\sin\omega_n t\right)$$

$$\therefore x = \begin{cases} \dfrac{F_0}{m\omega_n}\left(1-\dfrac{t}{t_0}-\cos\omega_n t + \dfrac{1}{\omega_n t_0}\sin\omega_n t\right) & (0 < t \leq t_0) \\[2ex] \dfrac{F_0}{m\omega_n}\left(-\cos\omega_n t - \dfrac{1}{\omega_n t_0}\sin\omega_n(t-t_0) + \dfrac{1}{\omega_n t_0}\sin\omega_n t\right) & (t_0 < t) \end{cases}$$